Chapter 6
Transport Planning Overview

Introduction

For many companies, transport related costs form a significant portion of the total distribution budget. If you increase the efficiency and reduce costs due to effective planning and management of the transport department, they can have a real effect on a company's profits.

To optimise the utilisation of transport resources and load fill, you need to make accurate information available to the transport department as early as possible in the delivery cycle. This allows efficient scheduling of deliveries in the short term, and assists strategic transport planning in the longer term.

Transport Planning provides a wide spectrum of facilities for planning and day-to-day management of transportation. Together with the appropriate Distribution applications, it provides the comprehensive solution for companies managing their own fleet and those using third party carriers or a combination of both.

General Concepts

You can make deliveries in four basic ways, using:

- Own transport facility
- Third party transport
- Parcel service
- Collection by customer.

Whichever method, or combination of methods you use, you take the following operational steps to plan the transport:

- Build loads
- Assign transport resource
- Manage movements
- Consign loads

Building Loads

You have to consider a number of factors when you are building a load. These factors fall into three categories: customer; product; and vehicle.

Customer Factors

- Geographical position
- Limits on vehicles and container types accepted
- Carriers accepted
- Days of the week
- Times of the day
- Authorisation required for delivery
- Exclusive deliveries.

Product Factors

- Temperature control
- Security.

Vehicle Factors

- Maximum capacity (weight, volume, number of containers)
- Driver restrictions (HGV required.)
- Availability of vehicle and driver.

You need to continuously view the forward delivery load. The forward view should present the requirements in the appropriate units (weight, volume, number of containers) by Transport centre, date, time, route, and/or vehicle type restriction.

Typically, the day before despatch (or same day) you finalise the load, with the vehicle and carrier assigned. You then issue the picking and preparation documents.

Assign Transport Resource

The procedure for assigning the transport resource differs for companies using their own transport from those using third party.

Using third party, you must notify the carrier of:

- Vehicle requirements
- Loading date, time and place
- Drops
- Required delivery dates and times
- Weights/volume/number of containers per load
- Product details.

You can then agree the rates as part of the confirmation. Typically, this is a specific confirmation of a long standing agreement, based on rate tables supplied by the carrier. This allows you to accurately calculate and accrue transport costs, and match costs to carrier invoices.

If you are using your own transport facilities, determine the requirements and assign a vehicle and driver. You should continuously review the despatch requirements to maximise utilisation and minimise the number of part loads sent out.

You often need to assign more than one carrier. For example if you use a master load to send a complete load to a destination, where it is split into sub-loads. Then use local carriers to deliver these sub-loads on smaller vehicles.

Manage Movements

This includes all of the tasks necessary to get the goods on the right vehicle at the right time and in the most efficient manner. You can use Transport Planning to produce the following paper work:

- Picking notes
- Despatch notes
- Load sheets
- Bills of lading (consignment notes)
- Invoices

The important fact is that you pick the load as a consignment. You can pick by order or by carrier or route or individual load and then marshall the load.

There can be different requirements for confirmation of the different movements, but normally document confirmation with exception is used. In the case of despatch notes, you can print a clean document before consigning the load to the carrier.

Consign Loads

Once loaded, you confirm the following transport details before despatch:

- Carrier used.
- Weight or volume of load and number of containers.
- Date and time of despatch.
- Vehicle and driver details - such as registrations or employee number.
- Driver instructions.
- Security details - including tag number, type and colour assigned to the vehicle.
- Thermostat settings (for chilled and frozen compartments).

The last three are particularly relevant for internal transport.

The driver normally sign a copy of the bill of lading to accept the load, that is the load has been consigned to the carrier.

Relationship To Other Modules

Transport Planning operates under the control of System Manager, and is an advanced module which extends the functionality of base modules.

To use Transport Planning, you must have:

- Inventory Management
- Sales Order Processing

You can use Transport Planning with:

- Advanced Order Entry
- Distribution Requirements Planning (DRP) - You use the DRP interface to plan transport for depot replenishment as well as for retail deliveries.
- EDI Application Interface - You use the EDI interface to electronically receive sales orders and make them available for transport planning.
- Telesales
- Warehouse Management - You can produce a consolidated pick per load (or route or carrier).
- Accounts Payable - Once authorised, you can automatically transfer carrier invoices to the Accounts Payable invoice log.
- Advanced Financial Integrator (AFI) - Advanced Financial Integrator can post carrier charges and payment variances into the General Ledger, based on user defined rules.

Transport Planning Configuration

You can operate Transport Planning for a number of companies.

Within each transport company, you set up transport centres by assigning a number of Inventory stockrooms. You can then manage transport on behalf of a number of physical depots.

The transport planner has a view of the transport requirements for each defined stockroom, but can build loads on behalf of each or amalgamate them onto combined loads. The former is for shipping direct from the depots and the latter for transhipping prior to despatching from a single delivery depot.

You can associate the transport centre with multiple sales companies, so the planner can manage transport across commercial companies or divisions within an organisation.

The AFI interface can post the calculated carrier freight charges and actual payment variances to multiple General Ledger companies.

Maintenance Of Set Up Data

As this is an advanced application module, you set up much of the Transport Planning data as part of the base modules. However, you must enter additional data for the maximum benefit.

Transport centres

A transport centre is a delivery depot, where you plan, pick and despatch transport loads. You maintain a number of control flags that affect the processing of loads:

- Add Orders Immediately To Load - If set to yes and Transport Planning finds a load with the same route, date, transport centre, and vehicle type, with sufficient capacity, it automatically adds the order to this load. If set to no and Transport Planning finds a suitable load, it suggests this to the planner. If Transport Planning does not find a suitable load, you must create a suitable load and add the order to this load, using the load build screen.
- Release Loads Before Goods Ready - Sales Order Processing maintains a total of orders per load and number of orders confirmed as despatched. If you set this flag to no, then Transport Planning will not release loads and print documentation unless these totals tally.
- Print Documentation Immediately After Load Released - If set to yes, then immediately after releasing a load, Transport Planning prints the load sheet, and bill of lading, if required. If set to no, the load prints in the next batch print run.
- Bypass Load Despatch - If set to no, you must confirm each load as despatched individually. This date and time stamps the load as it leaves the depot.
- Load Numbers - you can maintain sequential load numbers per transport centre with a user-defined prefix, or maintain a range of load numbers per transport company.

Transport planners

For security reasons you must define each Transport Planning user as a transport planner. You can authorise each planner to one or more transport centres. You allocate each planner a default centre. The centre description is shown at the top of all screens. A planner can change to any other transport centre to which you have authorised them.

You maintain the following control flags per planner:

- Release Load - determines if a planner can release loads for printing of documentation and processing.
- Override Capacity Errors - determines if the planner can build loads over the capacities defined to the vehicle type.
- Authorise Payments - determines if the planner can transfer carrier invoices to Accounts Payable.

- Authorise Adjustments - determines if the planner can transfer carrier invoices to Accounts Payable if there is a difference between the calculated charge and the payment amount.

Vehicle types

If you require capacity checking or analysis by vehicle type, you must maintain vehicle types.

For each vehicle type, you enter a code, description, and load capacity, by either maximum weight, or maximum volume.

If you assign a vehicle type to a load, Transport Planning outputs an error if you exceed any specified capacity.

Routes and Nodes

A route represents a journey between two geographical locations undertaken by at least one carrier (internal or external). You define the route by a code, description and, optionally, an origin and destination, which can be the same for circular routes. You can also specify, per route, if you require a bill of lading (BoL), its format and number of copies.

If you specify the origin and destination, you must define them as nodes, that is geographical start and end points for the route.

You can enter a default carrier and ship time and vehicle type for the route. When you create a load for this route, Transport Planning adds these defaults to the load.

Note: You associate the default route and drop sequence with delivery points using the Delivery Profile Maintenance option.

Carriers

You must enter for each carrier, internal or external, a code, name, and address.

You can also enter parameters to rate the load, that is calculate transport charges. If you have Accounts Payable installed, specify a supplier account to transfer a matched carrier invoice from Transport Planning onto the Accounts Payable invoice log.

If you want to check carrier route combinations, then define the routes each carrier covers, together with any route specific rating parameters.

Carrier rates

If you want Transport Planning to rate loads, that is calculate carrier freight charges, you must enter sets of rates and associate them with carriers. Each set of rates has a code and expiry date. You can set up future rates to use automatically when current rates expire.

There are four rating methods available:

1. A table of charges based on the load gross capacity, with a fixed rate per drop and a maximum and/or minimum drop charge. You can specify a surcharge value or percentage to accommodate unforeseen.
2. The same as for 1, except that the charge is based on the gross capacity of each drop.

3. A charge based on the gross capacity dropped into each area. For each area you enter the fixed charge per unit of capacity, the Surcharge value or percentage, the drop minimum and maximum charge and the area minimum charge
4. As for 3, except that you maintain the charge per area in a table based on gross capacity.

Note: The capacity can be gross weight, load volume, or number of containers.

Areas (zones)

You can define areas, by a code and description so that you can group load drops into areas (or zones). You assign the area code to the appropriate delivery points, then you can produce documentation and rate loads split by area.

Item transport details

You must maintain additional item details so Transport Planning can calculate the capacity requirements of each item. These are:

- Weight of the goods, expressed in a standard unit of weight, per selling unit
- Volume of the goods per selling unit.
- The number of issue units which fill a default container (pallet, case, and so on).
- The tare weight of the default container.
- The volume of the default container.

You must specify at least one of the capacities.

Delivery profiles

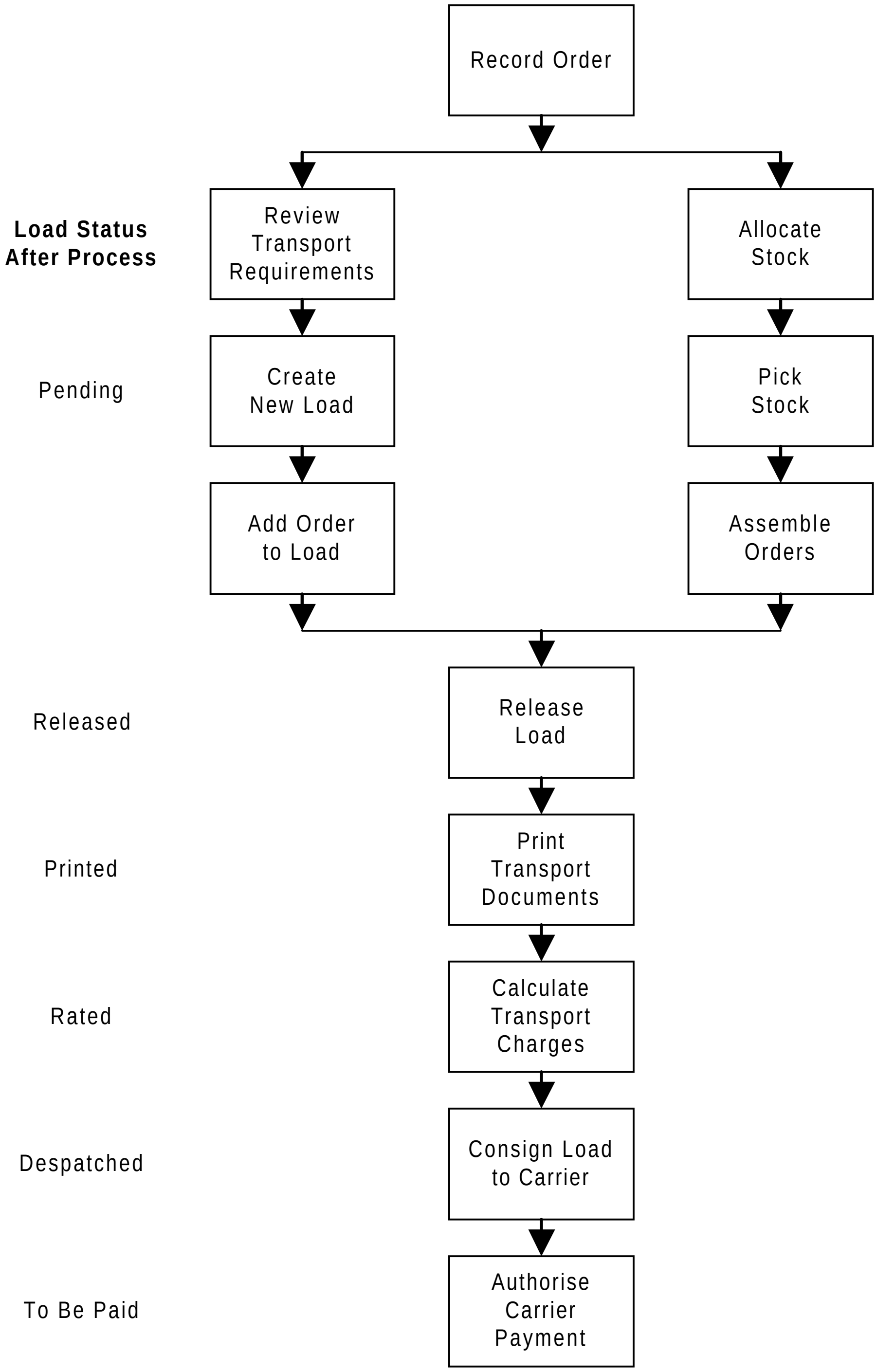
You must set up a delivery profile for each customer delivery point. The details needed are:

- Default Route, Drop and Carrier
- The area or zone used to split the drops for documentation and rating.
- If this load is exclusive to the customer, that is it cannot contain orders for other customers.
- If you must confirm the delivery at this delivery point requires.
- The days and times the customer accepts deliveries. (For memo purposes only).

You also maintain, for each transport company, when Transport Planning first creates transport requirements from Sales Order Processing. This can be either after you take the orders, when you allocate stock, when you create a picking note, or when you confirm the despatch.

Processing

The following diagram represents the business processes:



You can transfer the transport requirements of orders from Sales Order Processing immediately after taking the order, on allocation of stock, creation of pick notes or on confirmation of despatch. Any under or over pick or despatch confirmation is automatically reflected in Transport Planning.

You can automate the building of loads for specific transport centres. The transport planner only needs to create the pending load headers in advance and Transport Planning then adds the appropriate orders (up to any specified maximum capacities defined for the vehicle type.)

You can consign loads to the carrier manually, via the confirm load despatch screen, or automatically after printing and rating, where appropriate, the load.

You only need to match and authorise carrier invoices for rated loads.

Finally you close and copy the load to history, for future analysis.

Review Transport Requirements

In Transport Planning, the load build screen displays all the information necessary to assemble orders onto loads. The details displayed include:

- Gross weight or Volume or Number of containers.
- Sourcing stockroom.
- Default route and drop sequence.
- Default carrier and vehicle type.
- Ship date and time.
- Customer's name and account
- Suggested load number (if any).

You can use the following functions to display more details:

- Pending loads prompt (that is loads not yet released).
- Load details enquiry.
- Order enquiry.
- Delivery profile enquiry.

Orders with fixed or special instructions are highlighted with a flashing asterisk.

Also, an asterisk is displayed against the customer if there are any delivery restrictions.

Create New Load

You can create new load headers individually on line or altogether via a batch function.

Transport Planning can automatically generate load numbers or you can control them manually.

You must enter, as a minimum, a route and ship date.

You can set up skeleton load headers to represent a standard journey and then copy this each time you create a load.

If you have not set up a load header in advance, you can create it when you build a load.

Add Orders To Load

In the load build screen, you can select a number of orders to add to new or existing loads.

If you specify a vehicle type, then Transport Planning checks that its capacity is not exceeded.

You can move the order to alternative routes or resequence the drops or both.

When you add orders to loads, Transport Planning automatically removes them as requirements from the load build screen.

Release Load

Before you release a load, you can make any final amendments to drops, assign any driver, vehicle registration or security seal details via the load maintenance screen.

When you release a load, you can print the necessary transport documentation either immediately or next time you select a print run.

Once you release a load, it is no longer available for planning. You must return the status back to pending before you can amend any details.

Print Transport Documents

You can print load sheets, and optionally bills of lading, for each released load. These are described in detail in the Documents section of this overview.

Details of the route, drop and load print on each pick and despatch note, in Sales Order Processing. You can generate the notes for a specific carrier, route or load.

If Warehouse Management is installed, you can produce consolidated pick lists to efficiently pick for loading of vehicles.

Calculate Transport Charges

Transport Planning can calculate the expected carrier charge for selected loads.

The calculations are based on:

- Charge per capacity unit (gross weight, volume, containers).
- Penalty charge (If below specified minimum).
- Charge per drop.
- Minimum drop charge.
- Maximum drop charge.
- Surcharge value/percentage.
- Minimum charge per area.
- Maximum charge per area.
- Minimum charge per load.

You can maintain the charge rates (per unit or per drop) as a fixed amount or in a table based on capacity, that is weight, volume or number of containers.

There are four methods of automatic calculation:

- A charge based on the capacity, maintained in a table;
- A load charge based on the sum of drop charges which in turn are based on drop capacity, maintained in a table;
- A charge based on a fixed unit rate multiplied by the capacity dropped into each area (zone) contained in the load;
- As above except that the charge per area is based on a table of capacity ranges, rather than a fixed rate.

If Advanced Financial Integrator is installed, then you can post the calculated charges (prior to payment) into the General Ledger, so selected P & L and accrual accounts are debited and credited respectively.

Note: You can post to the General Ledger journals based on user defined attributes of the load, customer, product and carrier. The value may be apportioned over the products contained in a load, based on gross weight, volume or number of containers.

Consign Load To Carrier

You can confirm that each load is despatched. This makes sure that the load is date and time stamped when it leaves the depot.

You can bypass this stage, by allowing Transport Planning to automatically consign the load once the load sheet is printed.

Authorise Carrier Payment

When you receive a carrier invoice, you can match it against the calculated charge for one or more loads.

If there is a discrepancy, only authorised users can authorise the payment. Once authorised, Transport Planning automatically transfers the invoice and load details to the invoice log in Accounts Payable (if installed).

If Advanced Financial Integrator is installed then you can update the General Ledger with any variances between the rated and paid values.

Enquiries

You can group enquiries into three categories:

- Planning - What is to be done?
- Static information - What are the parameters?
- Status - Where are things?

Planning Enquiries

The Outstanding Loads Enquiry displays orders not included on loads.

If you select a range of dates, this shows the future outstanding requirements requiring planning of transport.

Static Information

You can view details for the following:

- Carriers
- Carrier route restrictions
- Delivery profiles
- Delivery time slots.

Status Enquiries

You can view the following load details:

- Pending loads (loads not yet released).
- Load details (load header information).
- Load drops (all drop details for selected load).
- Load orders (all orders for selected load).
- Load consignee (consignee details for selected load).
- Load totals (quantity/value totals for selected load).

You can use the Loads by Order enquiry to view the current status of an order, this includes which loads, if any, the order is assigned to. Select one of the loads to directly access the load enquiries.

Reports

Before you run a report, you can enter selection and sequencing parameters. All Transport Planning reports are defined to Report Generator so you can easily amend them.

Transport Planning prints the standard reports on standard listing paper with 132 print positions and 6 lines to the inch. The procedure or program number prints on the top left and the page number and date on the top right of the report.

Planning/Status Reports

The Load Status report provides a summary, per date or route, of the total capacity requirements of outstanding orders with a view of any current loads. Use this report to establish which routes are under or over committed before building loads.

The Outstanding Orders report provides details of all orders not included on loads, sequenced by route or drop within date. Use this report to view the detailed capacity requirements. This is particularly useful when you identify an over commitment on the Load Status report.

The Load Details report details all the drops and orders for selected loads.

Documents

You can print, and reprint:

- Load Sheets - print these for all released loads (including master loads). They list all drops for each load selected (in ascending or descending drop sequence). It serves three major purposes:
  - As a marshalling document for all despatch notes, especially if they are from more than one stockroom.
  - As an aid to loading the vehicle in the best sequence for dropping.
  - For the driver to use as an action and confirmation list.
- Bills Of Lading (BoL) - a more formal document than the load sheet, normally produced for selected external carriers. This can be in any one of four formats:
  - One BoL per load; One totals line; List of despatch note numbers.
  - One BoL per drop or delivery point; One totals line; List of despatch note numbers.
  - One BoL per load; One totals line per drop; List of despatch note numbers per drop.
  - One BoL per load; One totals line per area; List of all despatch note numbers on the load.

Financial Integrator Interfaces

The following functions relate to the data extraction via the AFI:

1. *Rating* - Manual or automatic rating creates charge details for loads that AFI can then extract for accrual accounting of transport costs.
2. *Payment* - Payment authorisation (manual or with Accounts Payable attached) updates charge details with payment information and transfers the invoice details to Accounts Payable. You can post any variances between rating and payment values to GL via AFI.
3. *Load Cancellation* - If you cancel a load that was rated but not extracted by the AFI, it creates a rating posting and a subsequent rating cancellation posting when you extract using AFI. Where you have extracted the rating posting by AFI, it only creates a rating cancellation posting. For loads you have not rated, no AFI extraction takes place.

Prerequisites

Applications

For Transport Planning to work correctly you must have the following installed:

- System Manager
- Inventory Management
- Sales Order Processing.

Transport Planning also interfaces with the following order capture modules:

- Advanced Order Entry
- Distribution Requirements Planning (DRP)
- EDI Application Interface
- Telesales
- Warehouse Management.

Transport Planning interfaces with the following financial modules:

- Accounts Payable
- General Ledger
- Advanced Financial Integrator.

Data

Items

You must define all items processed by Transport Planning to an associated sales or inventory stock holding company.

Customers

You must define all customer delivery points processed by Transport Planning to an associated sales company.

Transport Planning Configuration

You need to consider the following when you set up the data:

- Transport descriptions in Inventory
- Company profile
- Transport centres
- Planners
- Areas
- Routes and nodes
- Vehicle types
- Carriers
- Items
- Delivery profiles.

Transport Descriptions File Entries

You must define some entries to the Inventory descriptions file.

The following Description Identity Codes must be present for the Inventory company you use as the basis for the Transport Company:

TPAR

Adjustment reason.

TPBL

This is the bill of lading format (code length=2).

01 - One document per load.

02 - Document per drop/delivery point.

03 - One document with drop details.

04 - One document details by area.

TPCC

Commodity code.

TPCT

Carrier type.

TPCY

County.

TPDC

This is the capacity (code length=1).

1 - Weight.

2 - Volume.

3 - Containers.

TPIC

This when you can create transport requirements (code length=1).

1 - Order.

2 - Allocation.

3 - Pick.

4 - Commit (not used).

5 - Confirm pick (not used).

6 - Confirm despatch.

TPLS

This is the load status (code length=2).

10 - Pending (on hold, being planned).

20 - Aprinting (awaiting printing).

30 - Allocated (not used).

40 - On pick (not used).

50 - Picked (not used).

60 - Arating (awaiting rating).

70 - ACOD (awaiting confirmation of load despatch).

80 - Aauth.pay (awaiting payment authorisation).

90 - Aledg.post (awaiting posting of payment to the General Ledger).

95 - Ahist (awaiting drop to history).

99 - Cancelled.

TPPM

Freight payment method.

TPRM

This is the rating method (code length=2).

01 - Load and rate table.

02 - Drop and rate table.

03 - Area.

04 - Area and rate table.

TPSE

This is the serious errors (code length=2).

01 - Sales order not found.

02 - Pick note header not found.

03 - Item record not found.

04 - Stockroom balance not found.

05 - Route specified not found.

06 - Time out for requirements update.

07 - Details changed after AFI posting.

08 - Details changed after override.

09 - Global route change warning.

10 - Possible capacity problems.

11 - Item transport profile not found.

RSNC

Reason code.

TTYP

This is the text type (code length=1).

K - Carrier.

C - Customer.

O - Order.

T - Transport load.

<u>USGC</u>

This is the text usage (code length=1).

I - Internal.

P - Planner.

E - External.

*Note: The Text Type and Usage Combinations are:
KI - Internal carrier text
CP - Planner customer text
OE - External order text
TE - External transport load text.*

<u>WAIT</u>

This is the wait type for temporary record locking (code length=6).

BATCH - Batch lock request. Define BATCH entry with a limit of 5 for the record lock attempts and a rate of 20 for the wait seconds.

INTER - Interactive lock request. Define INTER entry with a limit of 5 for the record lock attempts and a rate of 2 for the wait seconds.

Note: The WAIT Description Identity controls how long a job waits for the release of a temporary record lock and how many times to retry to achieve a lock for updates.

Company Profile

The Transport company profile defines a number of primary defaults:

- When should the transport centre receive requirements from Sales Order Processing?
- Should transport load references be created at company or transport centre level?
- What are the significant decimal places for capacity types? (Weight, volume and number of containers.)
- What level do you plan orders, header or line detail? This determines if Transport Planning uses the stockroom code at header or line level to determine which the transport centre to place the order line requirements into.

Transport Centres

Transport centres define a delivery depot that process order requirements and loads. Consider the following:

- How many transport centres do you want to define for the company?
- How many inventory stockrooms do you want to associate with each centre?
- Should these stockrooms span sales companies, do you allow multiple company planning by a single centre?
- What is the default capacity type for the centre; weight, volume or number of containers?
- Should requirements be assigned to loads on initial creation, therefore bypassing manual load build and allowing Transport Planning to automatically add requirements to suitable loads?
- Should loads be available for release prior to confirming the despatch of all the orders assigned to the load?
- Should transport documentation be printed online when you release the load?
- Do you require confirm of load despatch?

Planners

You must define a planner profile for all users of Transport Planning. Consider the following issues:

- Can the planner plan loads across a number of transport centres?
- What level of authorisation should you assign to this planner. Can the planner:
  - Release loads?
  - Override capacity errors?
  - Authorise payments and adjustments?

Areas

You must define areas or zones so you can rate loads and produce appropriate documentation.

You should consider how these areas relate to a transport centre, that is, is it a geographical location or a radial pattern.

Routes And Nodes

You must define all routes together with optional start and end points. Consider the following issues:

- Are internal or external carriers associated with routes?
- Are vehicle types a factor, do certain vehicle types run on particular routes?
- Are origins and destinations applicable, is this between two locations rather than a circular trip back to the transport centre?
- Is formal documentation required for each route, do you need a bill of lading, or is a load sheet sufficient?

Vehicle Types

You only need to define vehicle types if you require analysis of transport requirements against vehicle capacities. Consider the following issues:

- Internal or external types or both (as defined by carriers).
- Are discrete weight and volume capacities important (these may be set as infinite requiring manual review/control of load capacities)?

Carrier Types

You must define all carriers. Consider the following issues:

- Is this internal transport or external carrier services.
- Is rating applicable?
- Are there any carrier route or rating restrictions?

Carrier Rates

You only need to define carrier rates if you want to calculate transport charges. This is generally only a feature of external transport of goods and rates are generally communicated on a contract basis via the carrier. Consider the following issues:

- Which of the rating methods is appropriate?
- How many rate codes and effectivity dates do you want to define?

Note: You may need to enter a large amount of data.

Items

You should review item details to assess capacity requirements for their transportation. Consider the following issues:

- Net weight and volumes per item issue unit.
- Tare weights (per issue unit or percentage of net weight) for any item wrappings.
- Quantity per default container.
- Container tare weight and volume.

Delivery Profiles

You must define all customer and distribution centre delivery points. Consider the following issues:

- Route and drop sequence.
- Default carrier.
- Does the customer require exclusive loads?
- Do you need to confirm delivery?
- Does the customer have specific times you can deliver?